Unit no. ° 15

CONSOLIDATION AND TRANSITION TO PREMIÈRE

25 May to 11 June 2027

I Review of calculation, functions and geometry

Example 15.1: Annual Review · Part A

1. Write 168 /252 in lowest terms; calculate 2 $^{-3} + 1/4$.

2. Factorise $x^2 - 6x$ + 9, then solve ($x - 3)^2 = 4$.

3. Solve $\dfrac{x-2}{x+1} \geq 0$.

4. For $f(x) = 4 - (x-1)^2$ on [−2; 3], give the maximum and the x-values where $f(x) = 0$.

5. With $A(0;1)$, $B(4;3)$, C(6; 4), prove that the points are collinear and give an equation of the line.

Solution:

1. $168/252 = 2/3$, and $2^{-3} + 1/4 = 1/8 + 2/8 = 3/8$. 2. $(x-3)^2$; $x - 3 = \pm 2$, so $S = \{1;5\}$. 3. Excluded -1, zero 2; signs + , −, +, so $S =]-\infty;-1[\cup[2;+\infty[$. 4. The square is positive or zero, so $f(x) \leq 4$, equality at x = 1. The zeros are $x = -1$, 3, both in the domain. 5. $\overrightarrow{AB} = (4;2)$, $\overrightarrow{AC} = (6;3) = \frac{3}{2}\overrightarrow{AB}$. An equation is $x - 2y + 2 = 0$.

Guidance for the teacher

Suggested mark scheme: 10 points, 2 per question. For question 3, reserve one point for the domain and the endpoints. For question 5, ask for the conclusion after the calculation.